



PAPER – 2

ADVANCED FINANCIAL MANAGEMENT



QUESTIONS

Advanced Capital Budgeting Decisions

1. ABC Ltd. plans to invest ₹ 16,00,000 in a new unit. The project is expected to have a useful life of 4 years, with no salvage value at the end of its life. The annual depreciation charge for the project is ₹ 400,000.

Projected revenues and costs for the project, ignoring inflation, are provided as follows:

Year	Revenues (₹)	Costs (₹)
1	12,00,000	6,00,000
2	14,00,000	8,00,000
3	16,00,000	8,00,000
4	16,00,000	8,00,000

ABC Ltd. is subject to a corporate tax rate of 60%, and the cost of capital for the project, including inflation premium, is 10%.

Depreciation provides a tax benefit, and inflation rates for revenues and costs over the project's lifespan are as follows:

Year	Revenue Inflation	Cost Inflation
1	10%	12%
2	9%	10%
3	8%	9%
4	7%	8%

Based on above information, answer the following questions:

- I. The depreciation tax benefit for the project per year shall be.....
 - (a) ₹ 300,000
 - (b) ₹ 240,000
 - (c) ₹ 360,000
 - (d) ₹ 400,000
- II. The inflation-adjusted revenue in Year 2 shall be.....
 - (a) ₹ 16,78,600
 - (b) ₹ 14,00,000
 - (c) ₹ 10,03,520
 - (d) ₹ 9,85,600
- III. The total cash inflow in Year 1 after adjusting for inflation and tax benefit on depreciation shall be.....
 - (a) ₹ 672,000
 - (b) ₹ 660,000
 - (c) ₹ 985,600
 - (d) ₹ 499,200
- IV. The inflation-adjusted cost in Year 2 shall be.....
 - (a) ₹ 16,78,600
 - (b) ₹ 14,00,000
 - (c) ₹ 10,03,520
 - (d) ₹ 9,85,600
- V. The present value of cash inflow for the year 3 shall be approximately.....
 - (a) ₹ 4,52,598
 - (b) ₹ 4,27,208
 - (c) ₹ 4,79,898
 - (d) ₹ 4,53,772

Financial Policy and Corporate strategy

2. In a recent Board Meeting of N Ltd. following financials of N Ltd. for the year ending 31st March 2025 were presented:

Balance Sheet as on 31.03.2025

		₹ '000	
Liabilities		Assets	
Equity Capital	4,80,000	Fixed Assets	2,42,000
10% Bonds	92,000	Cash	88,000
Sundry Creditors	66,000	Sundry Debtors	1,10,000
Bills Payable	88,000	Closing Stock	3,30,000
Other Current Liabilities	44,000		
Total Liabilities	7,70,000	Total Assets	7,70,000

Income Statement for the Year ending 31.03.2025

Particular	(₹ '000)	(₹ '000)
Sales		11,77,000
Less: Cost of Goods Sold		
Material	4,18,000	
Wages	2,64,000	
Factory Overheads	1,29,800	8,11,800
Gross Profit		3,65,200
Less: Selling & Distribution Cost	1,10,000	
Administrative Cost	1,22,800	2,32,800
Earnings Before Interest and Taxes (EBIT)		1,32,400
Less: Interest Charges		9,200
Earning Before Tax		1,23,200
Less: Taxes @ 50%		61,600
Net Profit (PAT)		61,600

During the Board Meeting:

- (i) Director A said that the company can maintain a certain growth even though the net profit margin remains constant, and assets increases proportionately to sales and it distributes its 30% of its net profit. To maintain this growth rate, it will not require any external funds.
- (ii) Director B proposed that just by maintaining a target capital structure and without issuing additional equity and maintaining target dividend pay-out ratio as proposed by Director A, more growth rate can be achieved.
- (iii) Director C though agreed with views of Director A and Director B, but is of the view that in the coming year it is expected that sales is likely to rise by 15%, hence if required we can go for issue of equity shares, bonds or debentures to achieve the same growth in sales.

From the information given above, choose the correct answer to the following questions:

- I. The Director A is talking about.....
 - (a) Internal Growth Rate
 - (b) Sustainable Growth Rate
 - (c) External Funding Requirements
 - (d) External Growth Rate
- II. The Director B is talking about.....
 - (a) Internal Growth Rate
 - (b) Sustainable Growth Rate
 - (c) External Funding Requirements
 - (d) External Growth Rate
- III. The Director C is talking about.....
 - (a) Internal Growth Rate
 - (b) Sustainable Growth Rate

- (c) External Funding Requirements
 - (d) External Growth Rate
- IV. If we go by the proposal of Director C, then approximately.....funds shall be raised from in form of equity or debt, assuming that dividend as proposed by Director A is paid out and assets and current liabilities are increased in the same proportion as increase in sales.
- (a) ₹ 1,15,500 thousand
 - (b) ₹ 85,800 thousand
 - (c) ₹ 79,332 thousand
 - (d) ₹ 36,212 thousand

Security Valuation

3. XYZ company has current earnings of ₹ 3 per share with 5,00,000 shares outstanding. The company plans to issue 40,000, 7% convertible preference shares of ₹ 50 each at par. The preference shares are convertible into 2 shares for each preference shares held. The equity share has a current market price of ₹ 21 per share.
- (i) What is preference share's conversion value?
 - (ii) What is conversion premium?
 - (iii) Assuming that total earnings remain the same, calculate the effect of the issue on the basic earning per share (a) before conversion (b) after conversion.
 - (iv) If profits after tax increases by ₹ 1 million what will be the basic EPS
 - (a) before conversion and
 - (b) on a fully diluted basis?
4. The following data are available for three bonds A, B and C. These bonds are used by a bond portfolio manager to fund an outflow scheduled in 6 year Current yield is 9%. All bonds have face value of ₹100 each and will be redeemed at par. Interest is payable annually.

Bond	Maturity (Years)	Coupon rate
A	10	10%
B	8	11%
C	5	9%

- (i) Calculate the duration of each bond.
- (ii) The bond portfolio manager has been asked to keep 45% of the portfolio money in Bond A. Calculate the percentage amount to be invested in bonds B and C that need to be purchased to immunise the portfolio.
- (iii) After the portfolio has been formulated, an interest rate change occurs, increasing the yield to 11%. The new duration of these bonds are: Bond A = 7.15 Years, Bond B = 6.03 Years and Bond C = 4.27 years.

Is the portfolio still immunized? Why or why not?

- (iv) Determine the new percentage of B and C bonds that are needed to immunize the portfolio. Bond A remaining at 45% of the portfolio.

Present values be used as follows :

Present Values	t_1	t_2	t_3	t_4	t_5
$PVIF_{0.09,t}$	0.917	0.842	0.772	0.708	0.650

Present Values	t_6	t_7	t_8	t_9	t_{10}
$PVIF_{0.09,t}$	0.596	0.547	0.502	0.460	0.422

Business Valuation

5. Equity of KGF Ltd. (KGFL) is ₹ 410 Crores, its debt, is worth ₹ 170 Crores. Printer Division segments value is attributable to 74%, which has an Asset Beta (β_p) of 1.45, balance value is applied on Spares and Consumables Division, which has an Asset Beta (β_{sc}) of 1.20 KGFL Debt beta (β_D) is 0.24.

You are required to calculate:

- (i) Equity Beta (β_E) of KGFL.
- (ii) Ascertain Equity Beta (β_E) of KGF Ltd., if it decides to change its Debt Equity position by raising further debt and buying back of equity to have its Debt Equity Ratio at 1.90. Assume that the present Debt Beta (β_{D1}) is 0.35 and any further funds raised by way of Debt will have a Beta (β_{D2}) of 0.40.
- (iii) Evaluate whether the new Equity Beta (β_E) justifies increase in the value of equity on account of leverage?

Derivatives Analysis & Valuation

6. (i) A Rice Trader has planned to sell 22000 kg of Rice after 3 months from now. The spot price of the Rice is ₹ 60 per kg and 3 months Futures on the same is trading at ₹ 59 per kg. Size of the contract is 1000 kg.

Required:

- (a) What the trader can do to mitigate its risk of reduced profit if the price is expected to fall as low as ₹ 56 per kg, 3 months hence.
- (b) Suppose if trader decides to make use of Futures, what would be the effective realized price from its sale when after 3 months, spot price is ₹ 57 per kg and Futures contract price for 3 months is ₹ 58 per kg?
- (ii) A company is long on 10 MT of copper @ ₹ 534 per kg (spot) and intends to remain so for the ensuing quarter. The variance of change in its spot and Futures prices are 16% and 36% respectively, having correlation coefficient of 0.75. The contract size of one contract is 1,000 kgs.

Required:

- (a) Calculate the Optimal Hedge Ratio for perfect hedging in Futures Market.

- (b) Advice the position to be taken in Futures Market for perfect hedging.
- (c) Determine the number and the amount of the copper Futures to achieve a perfect hedge.

Interest Rate Risk Management

7. Electra space is consumer electronics wholesaler. The business of the firm is highly seasonal in nature. In 6 months of a year, firm has a huge cash deposits and especially near Christmas time and other 6 months firm cash crunch, leading to borrowing of money to cover up its exposures for running the business.

It is expected that firm shall borrow a sum of €50 million for the entire period of slack season in about 3 months.

A Bank has given the following quotations:

Spot	5.50% - 5.75%
3 × 6 FRA	5.59% - 5.82%
3 × 9 FRA	5.64% - 5.94%

3-month €50,000 Futures contract maturing in a period of 3 months is quoted at 94.15 (5.85%).

You are required to determine:

- (a) How an FRA, shall be useful if the actual interest rate after 3 months turnout to be:
 - (i) 4.50%
 - (ii) 6.50%
- (b) How 3 months Futures contract shall be useful for company if actual interest rate turns out as mentioned in part (a) above.

Portfolio Management

8. Mr. A, a HNI invested on 1.4.2014 in certain equity shares as below:

Name of Co.	No. of shares	Cost (₹)
X Ltd.	1,00,000 (₹ 100 each)	2,00,00,000
Y Ltd.	50,000 (₹ 10 each)	1,50,00,000

In September 2014, 10% dividend was paid out by X Ltd. and in October 2014, 30% dividend paid out by Y Ltd. On 31.3.2015 market quotations showed a value of ₹ 220 and ₹ 290 per share for X Ltd. and Y Ltd. respectively.

On 1.4.2015, a technical analyst indicated as follows:

- (1) that the probabilities of dividends from X Ltd. and Y Ltd. for the year ending 31.3.2016 are as below:

Probability factor	Dividend from X Ltd. (%)	Dividend from Y Ltd. (%)
0.2	10	15
0.3	15	20
0.5	20	35

- (2) that the probabilities of market quotations on 31.3.2016 are as below:

Probability factor	Price/share of X Ltd.	Price/share of Y Ltd.
0.2	220	290
0.5	250	310
0.3	280	330

You are required to:

- Analyze the average return from the portfolio for the year ended 31.3.2015;
- Analyze the expected average return from the portfolio for the year 2015-16; and
- Advise Mr. A, of the comparative risk in the two investments.

Foreign Exchange Exposure and Risk Management

9. On 1st February 2025, XYZ Ltd. a laptop manufacturer imported a particular type of Memory Chips from SKH Semiconductor of South Korea. The payment is due in one month from the date of Invoice, amounting to 1190 Million South Korean Won (SKW). Following Spot Exchange Rates (1st February) are quoted in two different markets:

USD/ INR	85.00/ 85.50	in Mumbai
USD/ SKW	1390.00/ 1390.90	in New York

Since hedging of Foreign Exchange Risk was part of company's strategic policy and no contract for hedging in SKW was available at any in-shore market, it approached an off-shore Non-Deliverable Forward (NDF) Market for hedging the same risk.

In NDF Market a dealer quoted one-month USD/ SKW at 1390.00/1390.60 to be settled at reference rate declared by Bank of Korea.

After 1 month (1st March 2020) the dealer agreed for SKW 1385/ USD as rate for settlement and on the same day the Spot Rates in the above markets were as follows:

USD/ INR	85.50/ 85.75	in Mumbai
USD/ SKW	1388.00/ 1388.60	in New York

Analyze the position of company under each of the following cases, comparing with Spot Position of 1st February:

- (i) Do Nothing.
- (ii) Opting for NDF Contract.

Note: Both ₹/ SKW Rate and final payment (to be computed in ₹ Lakh) to be rounded off upto 4 decimal points.

10. Shanti exported 400 pieces of a designer jewellery to USA at \$ 400 each. To manufacture and design this jewellery she imported raw material from Japan of the cost of JP¥ 12000 for each piece.

The labour cost and variable overhead incurred in producing each piece of jewellery are ₹ 2,600 and ₹ 1300 respectively.

Suppose Spot Rates are:

$$\begin{aligned} \text{₹/ US\$} & \quad \text{₹ 80.00} - \text{₹ 81.00} \\ \text{JP¥/ US\$} & \quad \text{JP¥ 135} - \text{JP¥ 140} \end{aligned}$$

Shanti is expecting that by the time the export remittance is received and payment of import is made the expected Spot Rates are likely to be as follows:

₹/ US\$ ₹ 83.90 – ₹ 84.25

JP¥/ US\$ JP¥ 125 – JP¥ 131

You are required to calculate the resultant transaction exposure.

Mergers, Acquisitions and Corporate Restructuring

11. C Ltd. and P Ltd. both companies operating in the same industry decided to merge and form a new entity S Ltd. The relevant financial details of the two companies prior to merger announcement are as follows:

	C Ltd.	P Ltd.
Annual Earnings after Tax (₹ lakh)	10000	5800
No. Shares Outstanding (lakh)	4000	1000
PE Ratio (No. of Times)	8	10

The merger will be affected by means of stock swap (exchange) of 3 shares of C Ltd. for 1 share of P Ltd.

After the merger it is expected that due to synergy effects, Annual Earnings (Post Tax) are expected to be 8% higher than sum of the earnings of the two companies individually. Further, it is expected that P/E Ratio of S Ltd. shall be average of P/E Ratios of two companies before the merger.

You are required to determine the extent to which shareholders of P Ltd. will be benefitted per share from the proposed merger.

Mutual Funds

12. A mutual fund raised ₹ 150 lakhs on April 1, 2018 by issue of 15 lakh units at ₹ 10 per unit. The fund invested in several capital market instruments to build a portfolio of ₹ 140 lakhs, Initial expenses amounted to ₹ 8 lakhs. During the month of April, the fund sold certain instruments costing ₹ 44.75 lakhs for ₹ 47 lakhs and used the proceeds to purchase certain other securities for ₹ 41.60 lakhs. The fund

management expenses for the month amounted to ₹ 6 lakhs of which ₹ 50,000 was in arrears. The fund earned dividends amounting to ₹ 1.5 lakhs and it distributed 80% of the realized earnings. The market value of the portfolio on 30th April, 2018 was ₹ 147.85 lakhs.

An investor subscribed to 1000 units on April 1, 2018 and disposed it off at closing NAV on 30th April, 2018. Determine his annual rate of earnings.

International Financial Management

13. The Treasury desk of a global bank incorporated in UK wants to invest GBP 200 million on 1st January, 2019 for a period of 6 months and has the following options:

- (1) The Equity Trading desk in Japan wants to invest the entire GBP 200 million in high dividend yielding Japanese securities that would earn a dividend income of JPY 1,182 million. The dividends are declared and paid on 29th June. Post dividend, the securities are expected to quote at a 2% discount. The desk also plans to earn JPY 10 million on a stock borrow lending activity because of this investment. The securities are to be sold on June 29 with a T+1 settlement and the amount remitted back to the Treasury in London.
- (2) The Fixed Income desk of US proposed to invest the amount in 6 month G-Secs that provides a return of 5% p.a.

The exchange rates are as follows:

Currency Pair	1 Jan 2019 (Spot)	30 Jun 2019 (Forward)
GBP - JPY	148.0002	150.0000
GBP- USD	1.28000	1.30331

As a treasurer, advise the bank on the best investment option. What would be your decision from a risk perspective? You may ignore taxation.

Note: Calculate all figures in millions and round off them upto 4 decimal points.

Theoretical Questions

14. (a) What is a Unicorn startup? Who coined the term?
(b) Secondary participants play a vital role in strengthening securitization transactions. Explain.
15. (a) Briefly explain the various market indicators used in Technical Analysis.
(b) How the Buy and Sell signals are provided by Moving Average Analysis?

OR

Discuss the parameters used to identify Currency Risk.